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ADMN: BIT/2024/59761

COURSE: BIT4107 MOBILE APPLICATION DEVELOPMENT

DATE: 6/9/2026

TASK 1: DEVELOPMENT TOOLS ASSEMENT

1. The tools were already installed on my computer, so I did not need to reinstall them. I verified their availability by executing the appropriate commands in the Command Prompt (CMD).

Java Development kit

Version: java -version

Status: installed and working

```
C:\Users\user>java -version
openjdk version "17.0.14" 2025-01-21
OpenJDK Runtime Environment Temurin-17.0.14+7 (build 17.0.14+7)
OpenJDK 64-Bit Server VM Temurin-17.0.14+7 (build 17.0.14+7, mixed mode, sharing)

C:\Users\user>
```

2. Node .js

Version: node --version

Status: installed

```
C:\Users\user>node --version
v24.14.0
```

```
C:\Users\user>node --version
```

3. Git

Version: git --version

Status: installed

```
C:\Users\user> git --version
git version 2.47.0.windows.1

C:\Users\user>|
```

The other tools that I needed included

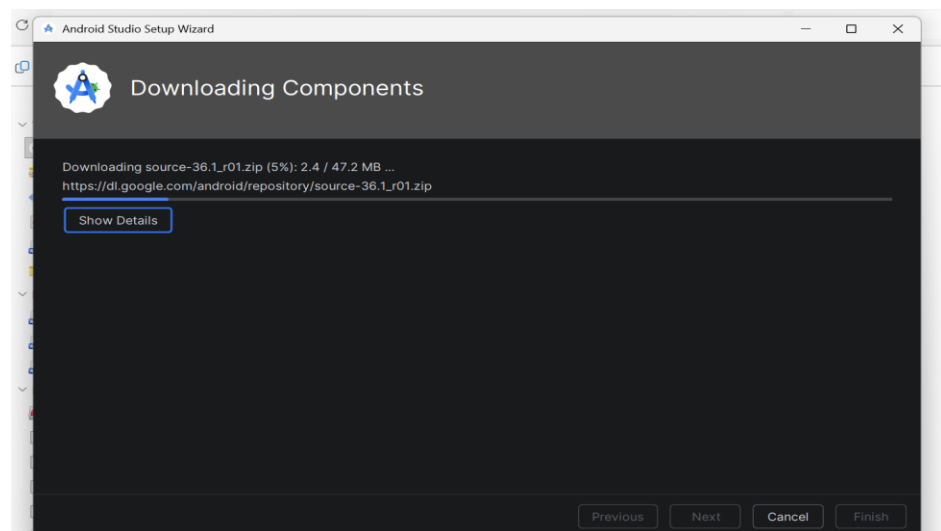
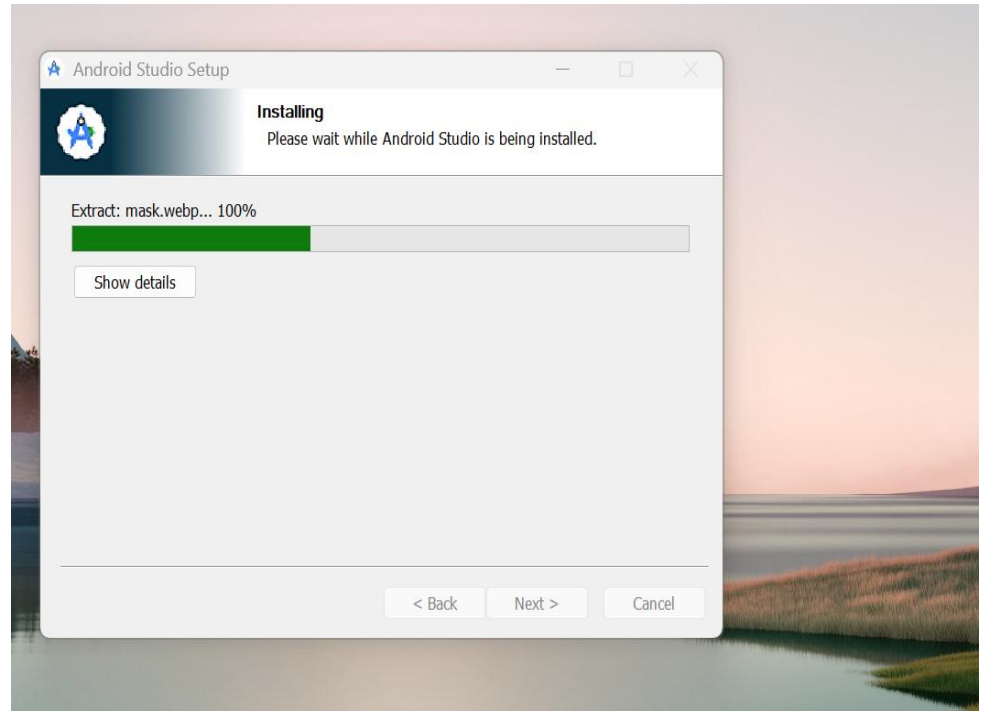
✓ Flutter sdk

```
project
analyze      Analyze the project's Dart code.
build        Build an executable app or install bundle.
clean        Delete the build/ and .dart_tool/ directories.
create       Create a new Flutter project.
drive        Builds and installs the app, and runs a Dart program that connects to the app, often to
              facing integration tests, such as with package:test and package:flutter_driver.
gen-l10n     Generate localizations for the current project.
pub          Commands for managing Flutter packages.
run          Run your Flutter app on an attached device.
test         Run Flutter unit tests for the current project.

Tools & Devices
attach       Attach to a running app.
custom-devices List, reset, add and delete custom devices.
devices      List all connected devices.
emulators    List, launch and create emulators.
install      Install a Flutter app on an attached device.
logs         Show log output for running Flutter apps.
screenshot   Take a screenshot from a connected device.
symbolize    Symbolize a stack trace from an AOT-compiled Flutter app.
widget-preview Manage the widget preview environment.

Run "flutter help <command>" for more information about a command.
Run "flutter help -v" for verbose help output, including less commonly used options.
```

✓ Android studio



✓ Python



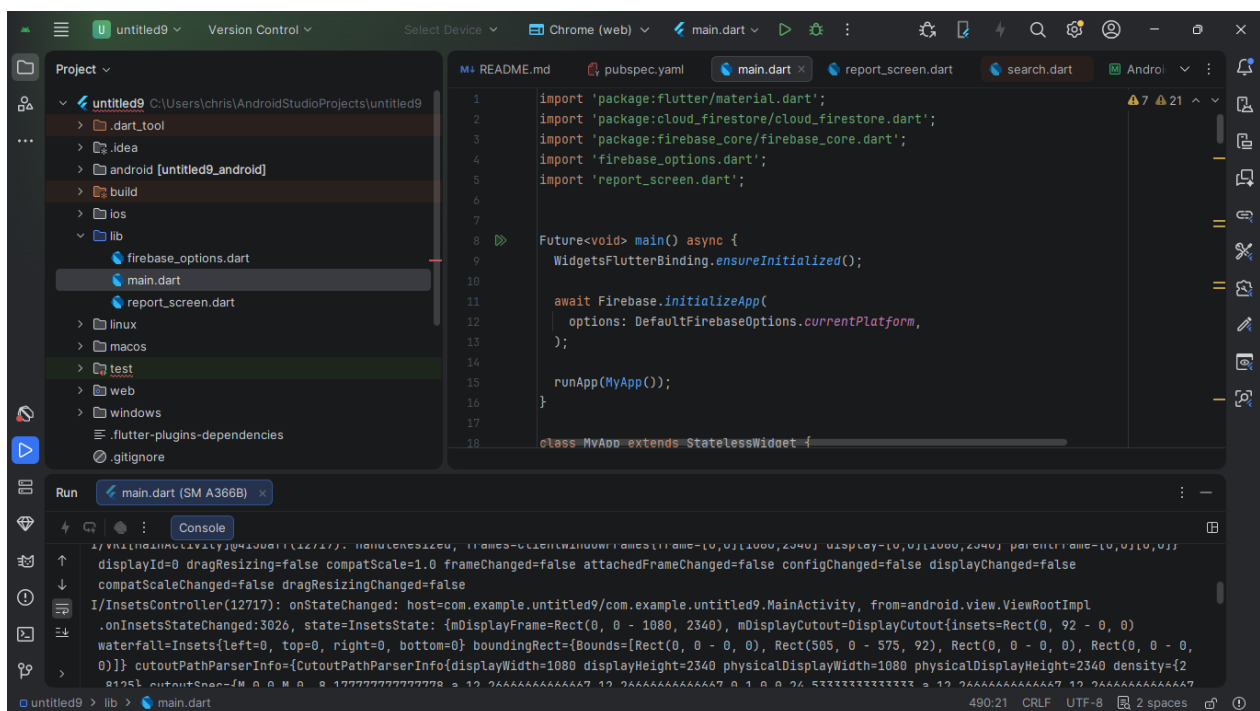
Which installed as shown above;

TASK 2. FIRST APP CREATION

App title: Student Registration App

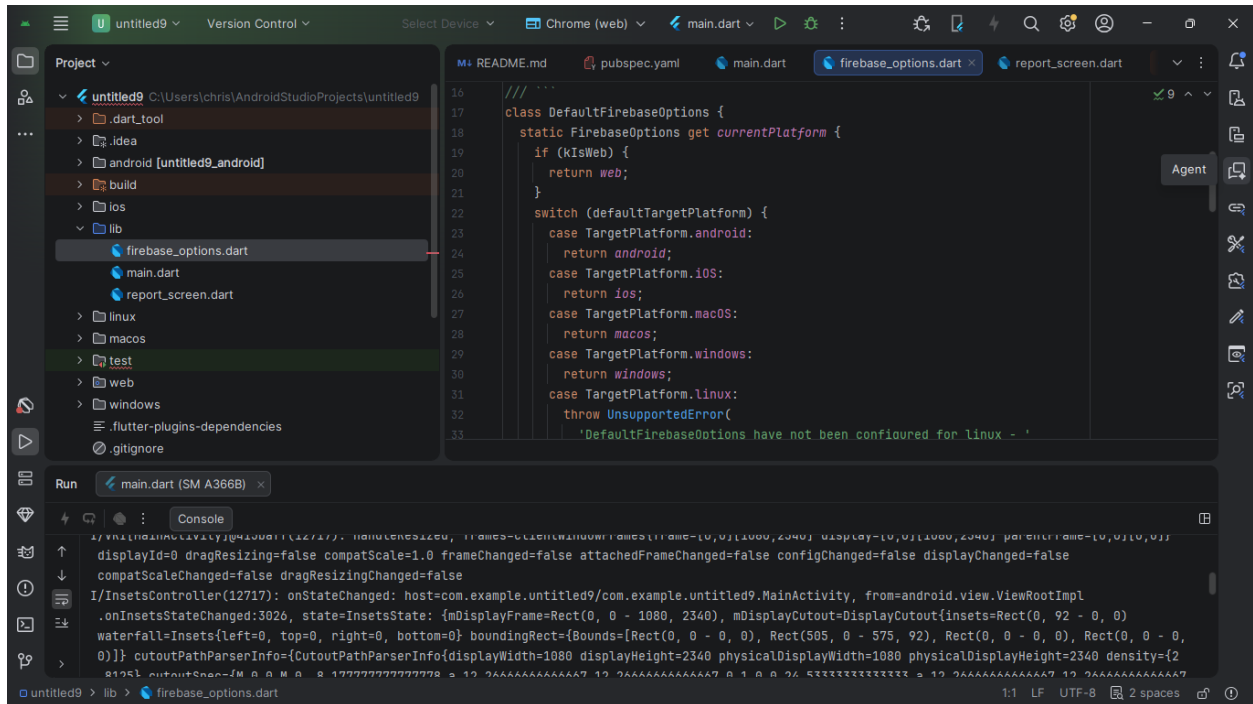
Background color: Maroon

Fontsize:14



Firebase Firestore Database Integration

I implemented a Firebase Firestore database as the primary data storage solution for the system. I selected Firebase due to its scalability, real-time synchronization capabilities, and ease of integration with Flutter applications. The database supports the storage, retrieval, updating, and deletion of records while ensuring data consistency across the platform.



🔥

Firestore

🔍

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Project Overview

⚙️

Settings

📄

What's new

Phone Verification

NEW

📄

Project shortcuts

Firestore

Realtime Database

SQL Connect

NEW

📄

Product categories

Databases & Storage

Security

Spark

No-cost (\$0/month)

Upgrade

NEW

Summit University

Cloud Firestore

Database (default)

>

students

>

5Fs0JivualqALPg1DorF

+ Start collection

+ Add document

+ Start collection

students

5Fs0JivualqALPg1DorF

Yt8vbqM7sNd7uxJ3p6db

Z1rFPgMjPZ1morIVAJVR

jfm8KiLGNuT56vs9SKoV

LoGHAcwbsvoYArHgIlv9

+ Add field

course: "Business"

createdAt: June 3, 2026 at 11:17:16 AM UTC+3

email: "sue@sue.com"

name: "Sue"

studentId: "78433"

year: "2nd Year"

Database location: europe-west2

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Firestore

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Project Overview

⚙️

Settings

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What's new

Phone Verification

NEW

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Project shortcuts

Firestore

Realtime Database

SQL Connect

NEW

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Product categories

Databases & Storage

Security

Spark

No-cost (\$0/month)

Upgrade

NEW

Summit University

Cloud Firestore

New in Cloud Firestore

Overview

Last 7 days

Jun 2 - Jun 9

Reads

206 total

Writes

26 total

Deletes

22 total

200

100

0

Jun 3

Jun 4

Jun 5

Jun 6

Jun 7

Jun 8

Databases

Upgrade to add database

Name	Edition	Location	Mode	Scheduled backups
(default)	Standard	europe-west2	Native	

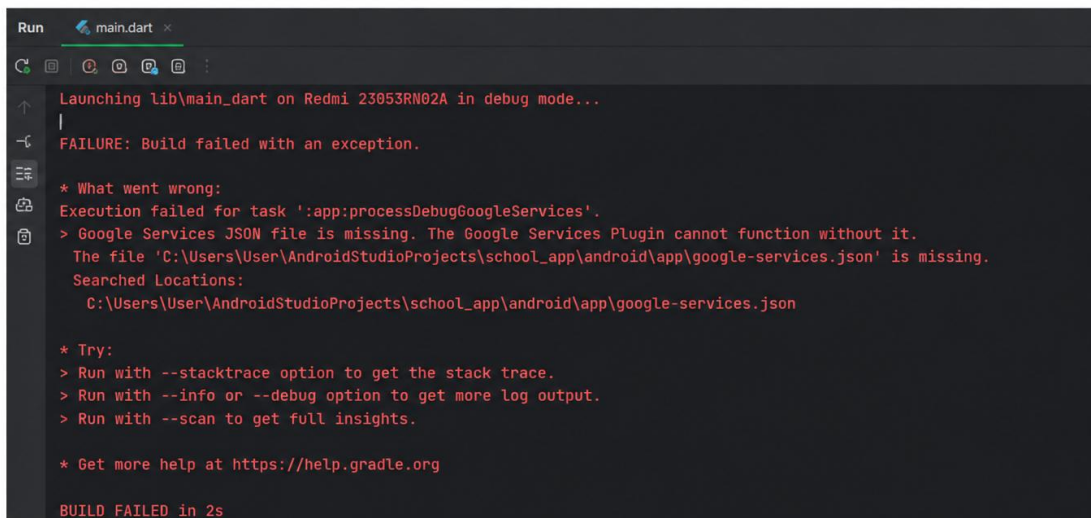
TASK 3: CHALLENGES AND SOLUTIONS

Challenge 1: Firebase Configuration and Connection Challenge

Problem: The Flutter application failed to connect to Firebase Firestore due to a missing or incorrectly configured google-services.json file, causing build errors and preventing database operations.

Trial: I reviewed the error messages, checked the Firebase configuration files, verified the project structure, and tested different configuration settings to identify the root cause of the issue.

Solution: I downloaded the correct google-services.json file from the Firebase Console, placed it in the app directory, and updated the required Firebase dependencies and Gradle configurations, which successfully established the connection between the application and Firebase Firestore.



```
Run main.dart x
Launching lib\main.dart on Redmi 23053RN02A in debug mode...
FAILURE: Build failed with an exception.

* What went wrong:
Execution failed for task ':app:processDebugGoogleServices'.
> Google Services JSON file is missing. The Google Services Plugin cannot function without it.
The file 'C:\Users\User\AndroidStudioProjects\school_app\android\app\google-services.json' is missing.
Searched Locations:
  C:\Users\User\AndroidStudioProjects\school_app\android\app\google-services.json

* Try:
> Run with --stacktrace option to get the stack trace.
> Run with --info or --debug option to get more log output.
> Run with --scan to get full insights.

* Get more help at https://help.gradle.org

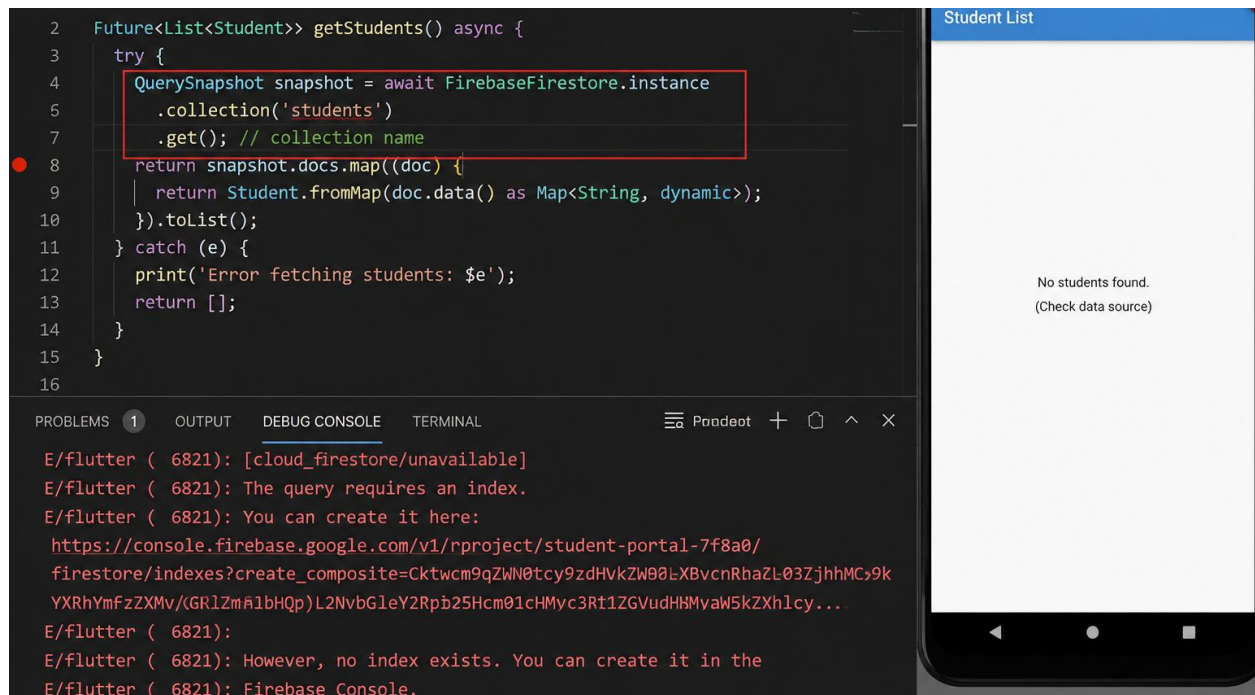
BUILD FAILED in 2s
```

Challenge 2.

Problem: The application experienced difficulties retrieving and displaying data from Firebase Firestore due to incorrect collection names and mismatched field names between the database and the Flutter code.

Trial: I examined the Firestore database structure, compared the collection and field names with those used in the application code, and tested multiple queries while monitoring the output and error messages.

Solution: I corrected the collection and field names in both Firebase Firestore and the Flutter application, ensuring consistency between the database schema and the code. This enabled successful retrieval and display of data within the system.

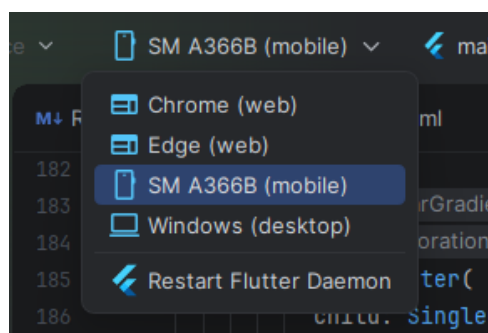


Challenge Three:

Problem: My emulator started malfunctioning and lagging on my PC.

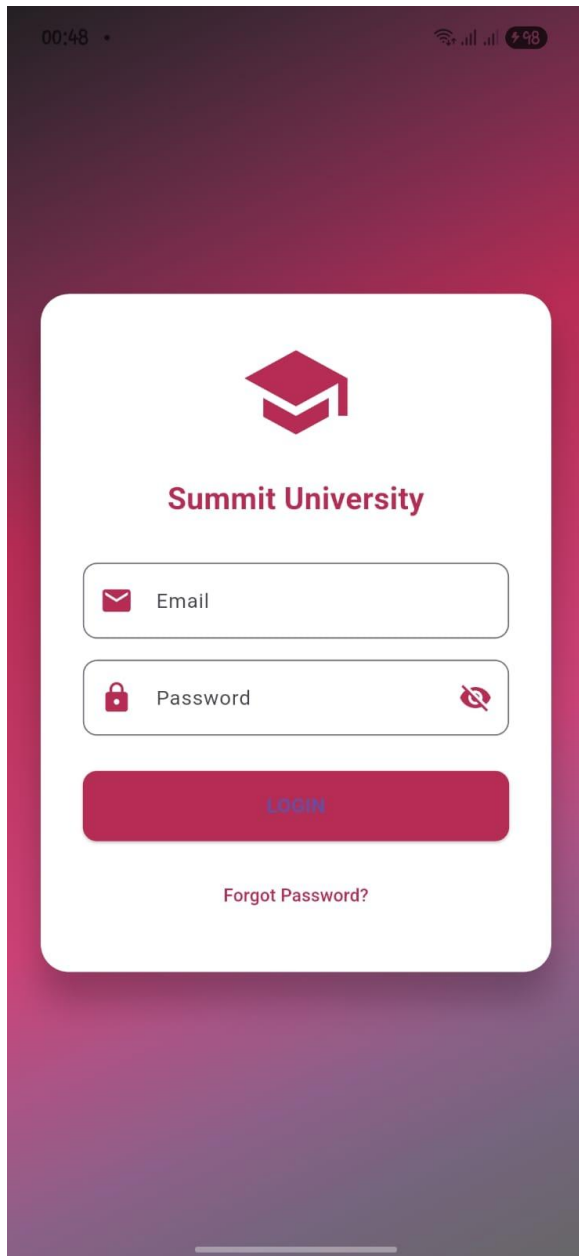
Trial: I freed some space on my PC to improve the PC performance and looked for an alternative

Solution: I connected my phone to my PC and used it as my emulator



WEEK2:STUDENT MINI MANAGEMENT

LOG IN Page



The image shows a mobile application login screen for Summit University. The background is a dark purple-to-pink gradient. A white rounded rectangle in the center contains the university's logo (a red graduation cap) and the name "Summit University" in red. Below the name are two input fields: "Email" with a red envelope icon and "Password" with a red padlock icon and a red eye icon for toggling visibility. A red "LOGIN" button is positioned below the password field. At the bottom of the white box is a "Forgot Password?" link. The top of the screen shows a status bar with the time "00:48", signal strength, and a battery level of "98".

00:48 98



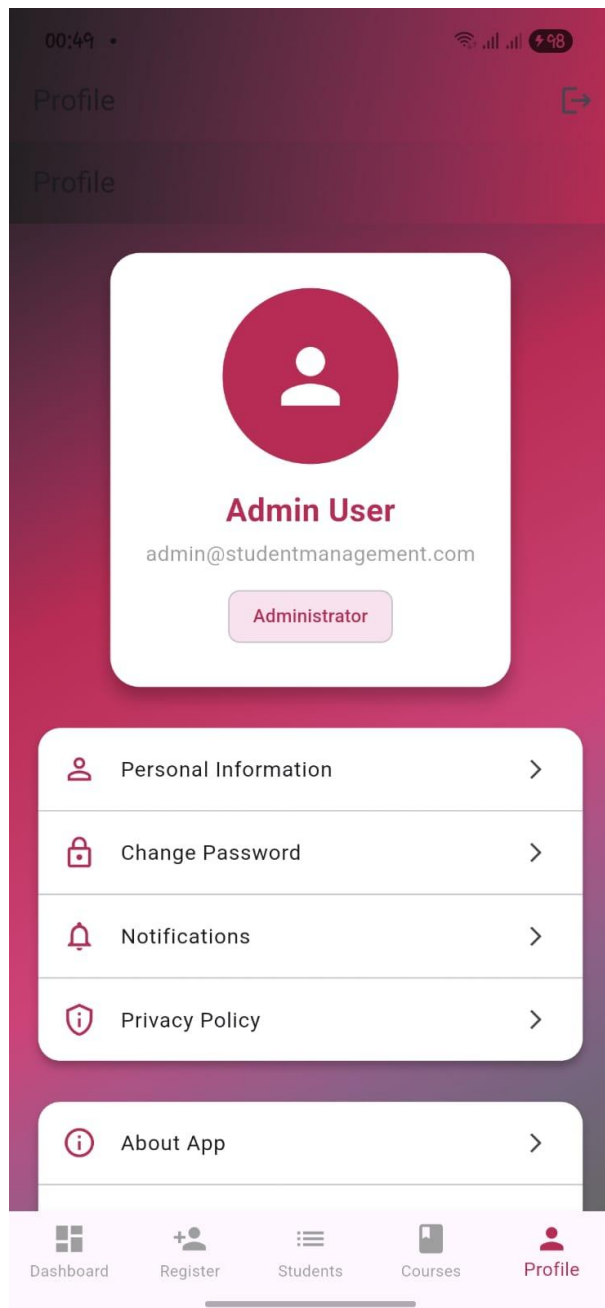
Summit University

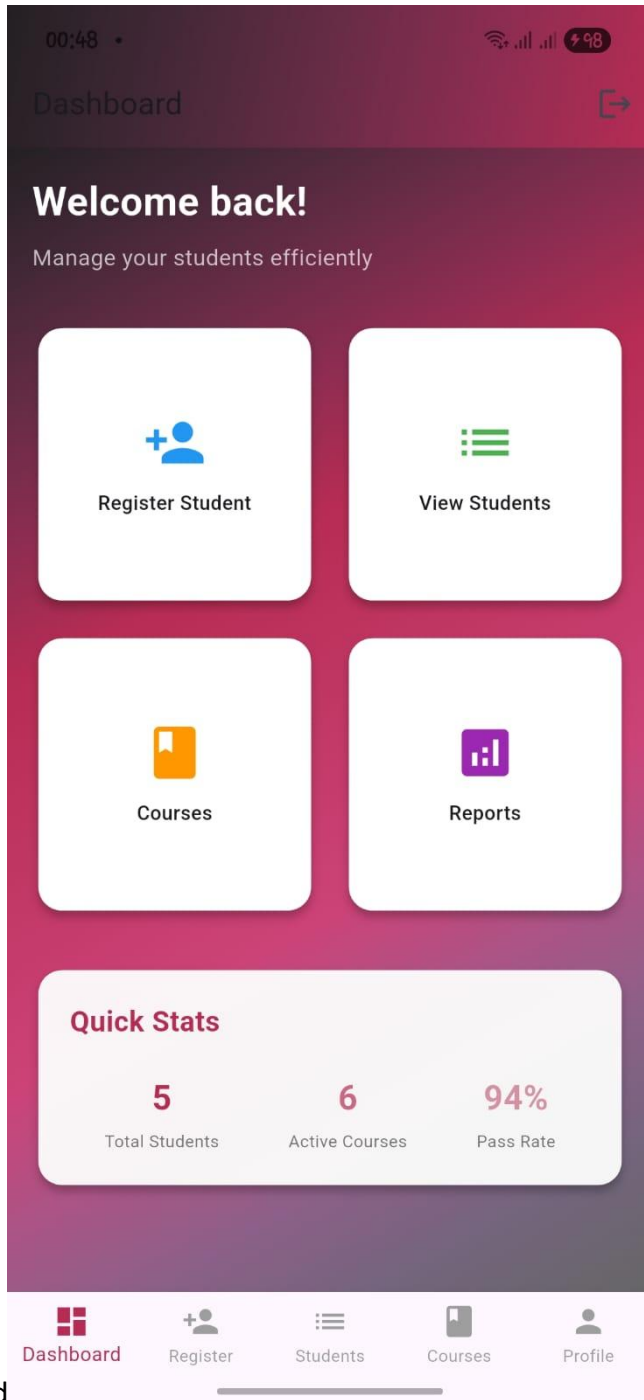
 Email

 Password 

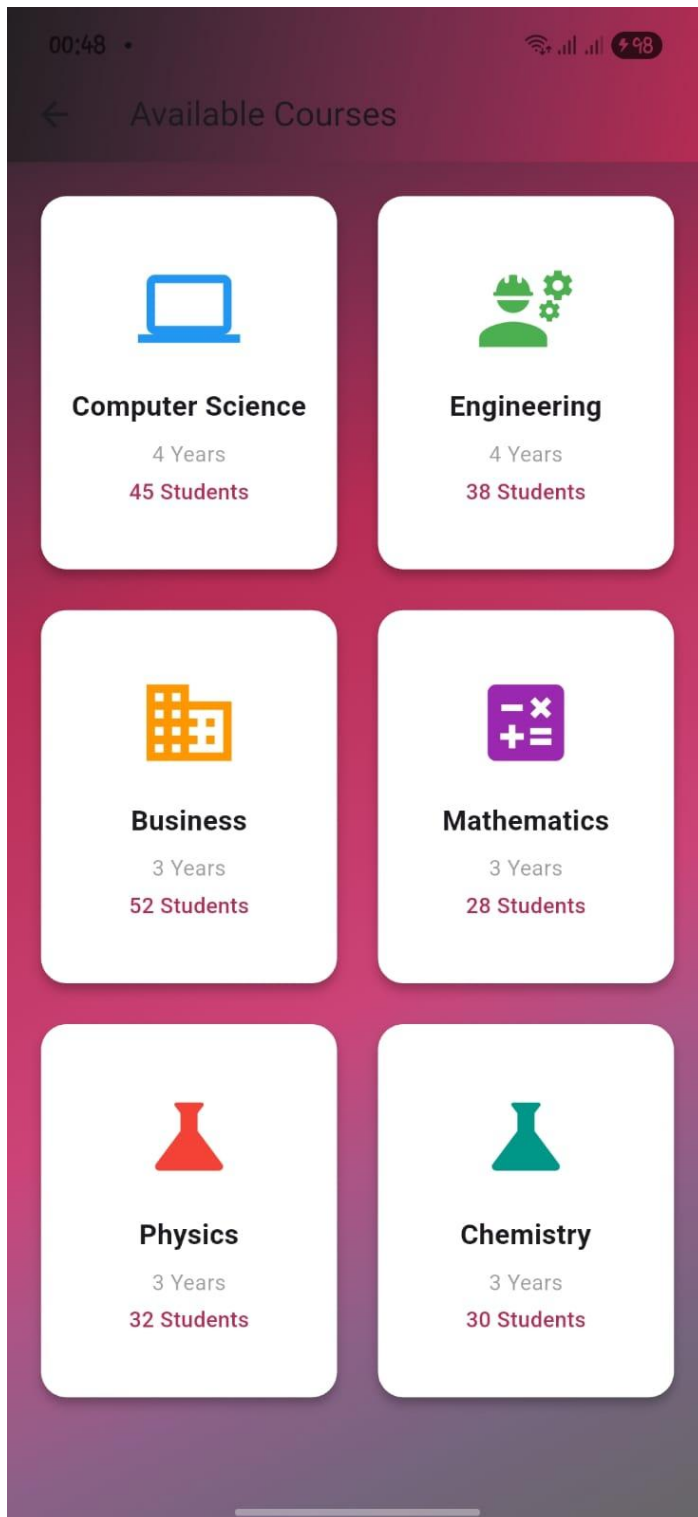
LOGIN

[Forgot Password?](#)





Dashboard



The available course are shown in this screen

00:50 •    98

Register Student 



Student Registration

 Full Name

 Student ID

 Email

Course

 Computer Science ▼

Year

 1st Year ▼

REGISTER STUDENT

 Dashboard

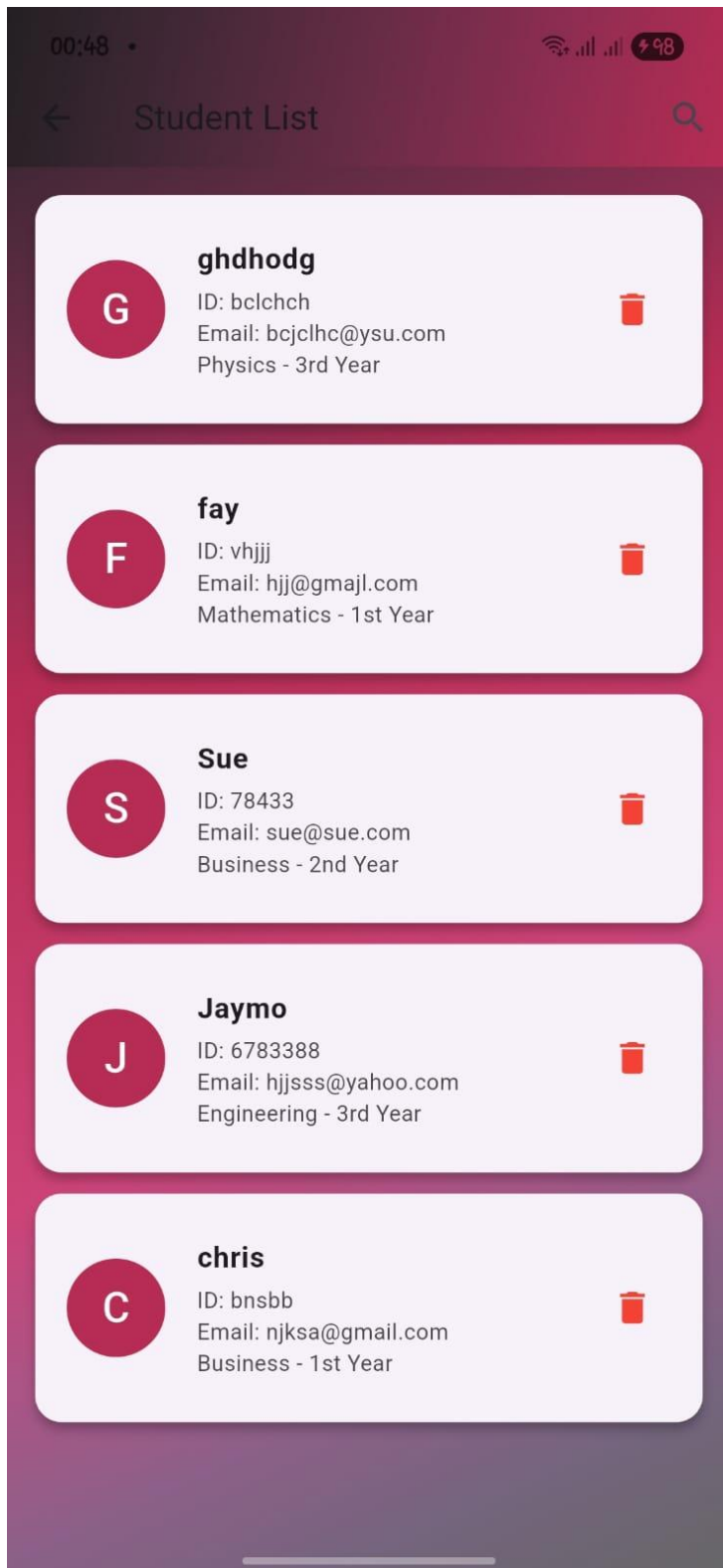
 Register

 Students

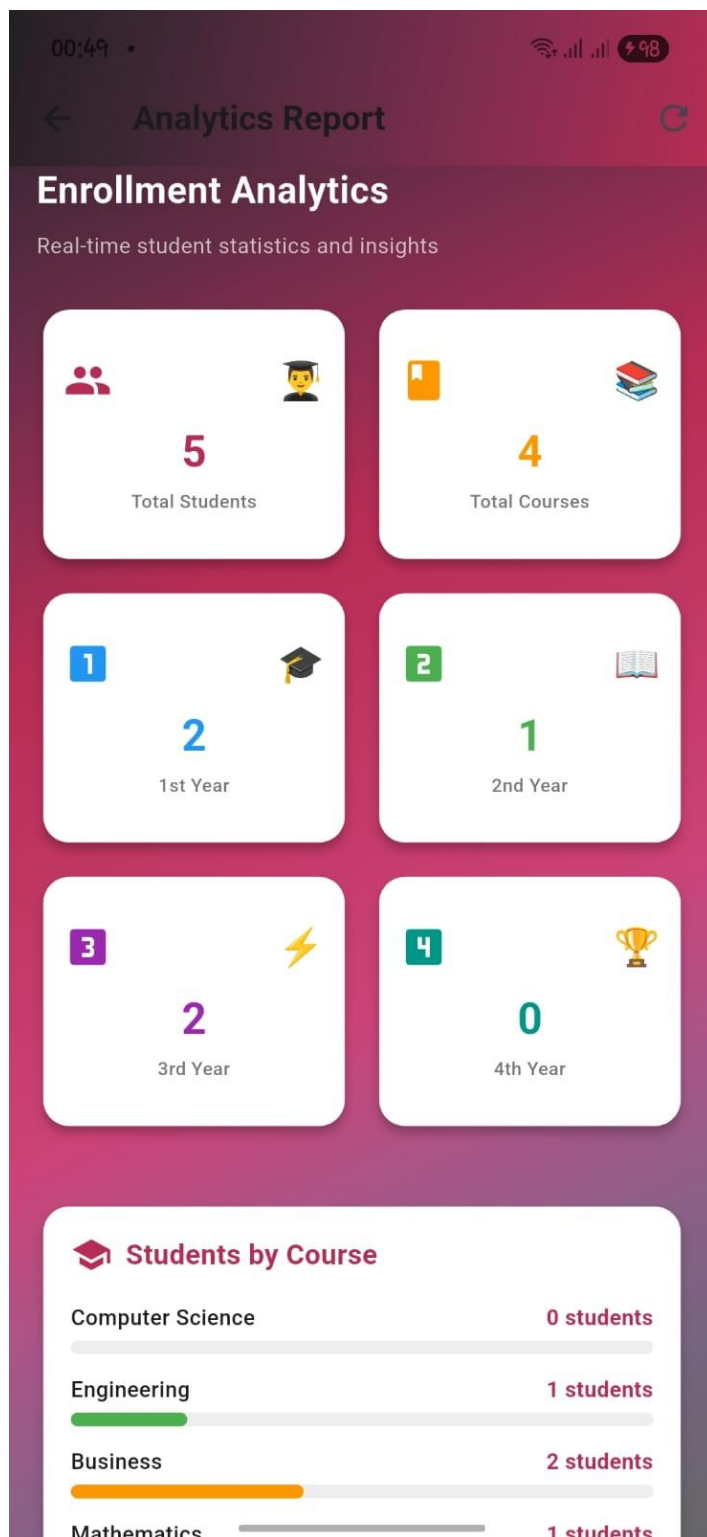
 Courses

 Profile

The registration screen with five screen at the bottom which are functional



The student list which is stored in the firebase database



This is the report screen

References

<https://firebase.google.com/docs> Google. (2026). *Firebase documentation*. Retrieved June 9, 2026, from Firebase Documentation.

<https://firebase.google.com/docs/firestore> Google. (2026). *Cloud Firestore documentation*. Retrieved June 9, 2026, from Cloud Firestore Documentation.

<https://firebase.flutter.dev>

FlutterFire. (2026). *FlutterFire documentation*. Retrieved June 9, 2026, from FlutterFire Documentation.

GitHub repository: GitHub link

Tools used: java 17, node.js 24, git 2.47